

Repository of Experimental Datasets for Spectroscopic Techniques

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Abstract. High-throughput metabolomics heavily relies on spectroscopic techniques, such as Infrared (IR), Raman, and Ultraviolet-Visible (UV-Vis) spectroscopy, to comprehensively characterize biological systems. However, despite their widespread application, the management of spectroscopic data significantly lags behind other analytical methods like mass spectrometry. Current data-sharing practices predominantly rely on general-purpose repositories that lack domain-specific metadata validation. This results in fragmented, "metadata-blind" datasets that fail to capture critical experimental contexts, thereby hindering cross-study integration and advanced machine learning applications. To address this critical infrastructure gap, this work presents the development and validation of a robust Extract, Transform and Load (ETL) pipeline, establishing the foundation for a centralized, domain-specific repository for spectroscopic datasets. The framework ensures that harmonized spectra adhere to FAIR (Findable, Accessible, Interoperable, and Reusable) principles. The proposed architecture was successfully validated against seven highly heterogeneous, open-access datasets. Statistical variability analysis confirmed that the automated data ingestion was seamlessly integrated with the specmine computational ecosystem while preserving the biological, physical, and chemical integrity of the raw spectra. This completed infrastructure standardizes spectroscopic data management, providing a highly curated foundation for integrative multi-omics analysis and advanced chemometric pipelines.

Keywords: Repository, IR, Raman, UV-Vis

1 Introduction

While initial research heavily focused on genomics, current paradigms emphasize integrative multi-omics approaches [1], critically including metabolomics to fully capture the dynamic complexity of biological systems. Within this framework, metabolomics has emerged as a crucial discipline. By providing comprehensive qualitative and quantitative data enabling the precise identification and chemical characterization of metabolites, it allows researchers to evaluate specific compounds of interest across diverse biological matrices and organisms [2]. However, the comprehensive characterization of these functional phenotypes relies entirely on

advanced analytical techniques, positioning spectroscopy as a fundamental tool for high-throughput metabolic profiling

Recent advancements in analytical chemistry have positioned spectroscopy as a fundamental, high-throughput tool for metabolic profiling. By analyzing the interaction between electromagnetic radiation and biological matrices, modern spectroscopic modalities, such as Infrared (IR), Raman, and Ultraviolet-Visible (UV-Vis) which enable rapid, non-destructive, and label-free biochemical characterization [3, 4].

However, despite their extensive application in scientific literature, the management of spectroscopic data significantly falls behind other analytical methods. Mass spectrometry (MS) benefits from well-established data standards and many robust centralized repositories, such as MetaboLights [5], MassIVE [6], PRIDE [7], NIST Mass Spectrometry Data Center [8], Metabolomics Workbench [9] and MassBank [10] which enforces strict curation of both raw data and experimental metadata. In contrast, the absence of standardized metadata in spectroscopic datasets (IR, Raman and UV-Vis) prevents the identification of molecular patterns across different omics layers. Although analyses limited to a single data type provide valuable insights, literature highlights that integrative multi-omics frameworks are significantly more effective in elucidating causative biological mechanisms [1]. Consequently, the fragmentation of spectroscopic data hinders the development of the integrative models required for a comprehensive understanding of biological systems.

To address these limitations, this work presents the development of a standardized infrastructure for management and integration of diverse spectroscopic datasets. The primary objective is to create a robust repository capable of consolidating biochemical and metabolomic data, such as Nuclear Magnetic Resonance (NMR), Gas chromatography–mass spectrometry (GC-MS), and Liquid chromatography–mass spectrometry (LC-MS) with spectroscopic modalities such as IR, Raman, and UV, particularly focused on complex matrices and natural products. This repository will implement data curation standards based on established reference databases like the, ensuring the inclusion of critical metadata. Furthermore, the infrastructure will be integrated with the specmine package [11] and the webspecmine application [12], providing a unified pipeline for the storage, retrieval, and analysis of multi-platform experimental data.

2 State of the art

2.1 Spectroscopic Techniques

Infrared (IR)

All organic compounds and many inorganic compounds absorb IR radiation [13]. The scanning of a molecular sample with successive infrared frequencies enables the extraction, qualitatively, of mechanical vibration frequencies of the molecule [14].

IR spectroscopy remains a cornerstone of analytical metabolomics due to several inherent strengths. First, the technique offers a significantly higher signal-to-noise ratio and sensitivity compared to Raman scattering, which is a fundamentally weaker physical process. This efficiency allows for rapid data acquisition, a critical factor in high throughput screening. Furthermore, IR spectroscopy benefits from extensive spectral libraries (e.g. NIST and Wiley), facilitating the identification of unknown compounds through established reference standards [15]. The adoption of Attenuated Total Reflectance (ATR) sampling has further enhanced the technique's utility, allowing for the direct analysis of diverse biological matrices with minimal preparation, thereby reducing experimental artifacts and improving reproducibility. Finally, IR is particularly sensitive to changes in the molecular dipole moment, providing superior detection of polar functional groups, such as carbonyls and amides, which are essential for characterizing protein secondary structures and lipid profiles in complex natural products [15,16].

A major bottleneck in Infrared (IR) spectroscopy for biomedical applications is the intense absorption of radiation by water molecules. Due to the high polarity of the OH bond, water exhibits dominant absorption bands which obscure critical biological signals such as the Amide I band of proteins [15].

Raman

Similarly to IR, Raman spectroscopy analyzes molecular vibrations; however, it is based on the inelastic scattering of monochromatic laser light upon interaction with molecules, resulting in a Raman shift. These frequency shifts correspond to the excitation of specific vibrational levels, providing detailed information regarding molecular structure and conformation [17]. A critical advantage of Raman spectroscopy over infrared techniques is its insensitivity to water interference, making it exceptionally suitable for analyzing biological samples with high water content in their native state [18,19].

Beyond its compatibility with aqueous environments, Raman spectroscopy offers superior spatial resolution, often reaching the sub-micron scale. This allows for "chemical imaging" or Raman microscopy, enabling the mapping of metabolite distribution within single cells or complex tissue structures without the need for extrinsic labels [20]. Furthermore, the use of confocal Raman microscopy allows for non-invasive optical sectioning, providing depth profiles of heterogeneous samples, which are particularly relevant for natural products and multilayered biological matrices [21]. The primary drawbacks of Raman spectroscopy is its inherent weakness, small number of photons undergoes Raman scattering. This often leads to low sensitivity and long acquisition times to achieve an acceptable signal-to-noise ratio [20]. Additionally, biological samples frequently exhibit strong autofluorescence, which can be several orders of magnitude more intense than the Raman signal, effectively hindering the spectral data information [22].

UV-Vis

Unlike vibrational techniques, UV-Vis spectroscopy measures the absorption of light resulting from electronic transitions within molecules, typically involving π -electrons or non-bonding electrons (n-electrons). This technique is fundamental for detecting conjugated systems, aromatic compounds, and certain metal complexes in biological matrices [23].

UV-Vis spectroscopy excels in its robustness for quantification, governed by the Beer-Lambert Law, which establishes a linear relationship between absorbance and analyte concentration. This makes it an essential tool for determining the concentration of proteins (e.g., at 280 nm) and nucleic acids (e.g., at 260 nm), as well as monitoring enzymatic reactions in real-time [24]. Furthermore, UV-Vis instrumentation is highly accessible, cost-effective, and easily integrated into automated pipelines, such as HPLC-coupled detectors, providing a high degree of reliability for routine laboratory analysis [25].

One of its main limitations of UV-Vis spectroscopy is its low molecular specificity. Because many different compounds can have broad, overlapping absorption bands in the UV-Vis range, it is often difficult to identify specific metabolites in complex mixtures without prior chromatographic separation [23]. Additionally, the technique is highly sensitive to the chemical environment, where factors such as pH, temperature, and solvent polarity can cause shifts in the absorption maximum potentially leading to inaccurate quantifications if these variables are not controlled [25].

2.2 Metadata Frameworks

The scientific utility of spectroscopic datasets is intrinsically linked to the rigor of their associated metadata. As highlighted by integrative multi-omics frameworks, the absence of standardized experimental context, such as sample preparation, instrument settings, and environmental variables renders raw spectra nearly impossible to replicate or integrate into larger cross-study analyses [1, 19].

2.2.1 Technique-Specific Requirements

Each spectroscopic modality demands a tailored metadata schema to account for their different necessary criteria. In the case of Raman, due to the high sensitivity of the Raman effect to the chemical environment, the documentation of pH levels is critical, as protonation states can significantly shift vibrational bands. Furthermore, the laser excitation wavelength and incident power must be recorded to account for varying fluorescence backgrounds and potential thermal degradation of biological samples [22].

Infrared (IR) Spectroscopy: Metadata must specify the sampling mode (e.g., ATR vs. transmission) and the physical state of the sample (e.g., hydrated tissue vs. lyophilized powder), as these factors dictate the effective path length and the degree of water interference [16].

UV-Vis Spectroscopy: Following the Beer-Lambert law, the optical path length and solvent polarity are mandatory fields. Without these, quantitative comparisons of molar absorptivity across different experimental setups are mathematically invalid [25].

2.2.2 Landscape of Existing Repositories and Data Acquisition

Despite the increasing volume of open-access spectral data, most general-purpose repositories, such as Zenodo [26] or Figshare [27], lack domain-specific metadata validation. In these platforms, most data is scattered in different formats, showcasing the lack of standard (e.g., proprietary vendor files or non-standardized CSVs). On the other hand, specialized multi-omics repositories like MetaboLights [5] focus predominantly on MS leaving a significant infrastructure gap for spectroscopy techniques. The proposed repository allows to fill this void by enforcing a metadata-first approach, ensuring that every stored spectrum is "FAIR" (Findable, Accessible, Interoperable, and Reusable) for both humans and machine learning algorithms.

Analyzing recent deposits on Zenodo reveals a wide variety of data, ranging from forensic analysis of hair dye colorants via Raman spectroscopy [28], near-infrared Raman analysis of redyed fabrics [29], materials science matrices such as battery interphase components via Attenuated total reflectance-Fourier transform infrared (FTIR) spectroscopy (ATR-FTIR) [30], to UV-Vis analyses of botanical extracts [31]. The absence of a centralized repository that categorizes these biological datasets according to their acquisition methodologies underscores the critical need for the domain-specific infrastructure developed in this project.

To address this challenge of data fragmentation and formatting, recent computational frameworks have begun to emerge. For instance, the Python-based library RamanSPy includes a dedicated datasets module that curates and standardizes programmatic access to a selection of valuable open-source Raman datasets, facilitating their direct use in machine learning workflows [32].

2.3 Integration with the specmine ecosystem

In the broader landscape of spectroscopic analysis, several robust computational tools have emerged within the R ecosystem. Notable examples include hyperSpec [33], which is highly optimized for managing hyperspectral imaging and Raman datasets, and ChemoSpec [34], which provides extensive tools for exploratory chemometrics (e.g., Principal Component Analysis (PCA) and Hierarchical Cluster Analysis (HCA)) across various spectroscopic modalities. Although, these packages typically operate as isolated local environments that require significant manual data wrangling.

To improve and unify the analysis of such data, in collaboration with UFSC in Brazil, the CEB's BiSBI research group has been developing the specmine framework [11]. This comprehensive, open-source R package and web-based application

(webspecmine) is designed for the chemometric and machine learning analysis of both multiomics and spectroscopic data. It equips researchers with a robust suite of analytical tools, which are broadly categorized into three domains: automated spectral preprocessing (e.g., baseline correction, smoothing, and peak alignment), exploratory data analysis (such as Principal Component Analysis and clustering), and predictive modeling (e.g. Partial Least Squares Discriminant Analysis and Random Forests). These diverse methods have been utilized to create various analytical pipelines for metabolomics (GC-MS and NMR) and spectroscopic (IR, UV-Vis) data. However, despite its advanced analytical pipeline, a significant constraint remains in the data ingestion phase. Currently, the specmine package demands manual work, relying heavily on the importation of local CSV files.

Data Harmonization

To rearrange and organize the data, a dedicated Extract, Transform, and Load (ETL) pipeline was conceptualized and implemented. The primary objective of this ETL process is data harmonization which converts diverse, unstructured formats and unstandardized metadata into a uniform, FAIR-compliant structure before ingestion into the repository.

A core architectural principle driving this development is the adoption of a minimal dependency framework. In the context of bioinformatics and software engineering, heavy reliance on third-party libraries often introduces structural vulnerabilities. In such scenarios, future updates, deprecations, or conflicts within external packages can inadvertently break the data pipeline. To mitigate these risks, the Extract, Transform, and Load (ETL) processes particularly the metadata parsers and harmonization scripts will be developed primarily using native, custom-built functions. By minimizing the reliance on external dependencies, other than the package *hyperspec* which was needed to ingest files in the *spc* format, this architectural choice ensures the longterm stability, reproducibility, and maintainability of the repository, guaranteeing its resilience as a robust backend for the *specmine* ecosystem.

A particular challenge arises when consolidating Raman spectroscopic datasets acquired across different instruments or experimental sessions, where the spectral axis (Raman shift, cm^{-1}) may differ in step size or range between individual sample files. Direct concatenation of such files into a single data matrix is not feasible without prior alignment of the spectral axis. To address this, a linear interpolation step was integrated into the concatenation function, mapping all individual spectra onto a common reference axis defined by the first sample in the set. This approach preserves the overall spectral profile of each sample while enabling the construction of a unified matrix.

The harmonized output is produced in two complementary formats. An RDS file preserves the complete specmine dataset object, including the spectral matrix, metadata table, x-axis labels, and modality type for direct ingestion by the specmine R package and its associated analytical pipelines. A CSV export is additionally generated for compatibility with the webspecmine web application.

3 Methods

3.1 Pipeline Architecture

The ETL pipeline is implemented in R and structured around a bipartite architecture, separating the harmonization process into two independent modules: a diagnostic parser (Parser 1) and an execution parser (Parser 2). Communication between the two modules is achieved through a named list of flag variables, initialised with default values and progressively populated during the diagnostic phase.

Parser 1 — Diagnostic Module

Parser 1 analyses the structural properties of the input file without performing any transformation on the data. Six diagnostic functions are executed sequentially, each responsible for detecting a single property and updating the corresponding flag in the shared variable list. The flags determined in this phase are file format and extension (`extensao_ficheiro`), column delimiter and decimal separator (`separador_colunas`, `separador_decimal`), number of metadata header lines (`linhas_metadados_topo`), matrix transposition state (`matriz_transposta`), header presence (`cabecalho_presente`), and x-axis order (`eixo_x_invertido`).

Delimiter detection is performed by sampling the first five lines of the file and counting occurrences of candidate separators. Metadata header detection operates by identifying the most frequent column count across the first 150 lines, treating any preceding lines as metadata. Transposition detection evaluates whether the spectral axis progresses monotonically along the first column or the first row, with a geometric fallback based on matrix dimensions. Header detection attempts to coerce the first data row to numeric values, inferring the presence of a header from the resulting conversion failures.

Parser 2 — Execution Module

Parser 2 receives the flag list produced by Parser 1 and executes the necessary transformations in sequence. Raw data extraction isolates the numeric matrix from the file by filtering lines that begin with a numeric character and contain the detected delimiter, discarding all metadata lines regardless of their position in the file. For binary .spc files, extraction is delegated to the `hyperSpec` package [37], wrapped in error and warning handlers to manage encoding incompatibilities. Matrix transposition, x-axis inversion, column name sanitization, and residual NA removal are applied sequentially based on the corresponding flags. Global metadata extraction employs a block-aware parser that traverses all lines in the file, distinguishing numeric rows from metadata rows using a heuristic based on line-initial characters and word length. Metadata blocks

are assigned positional prefixes (Bloco1_, Bloco2_, etc.) to prevent key overwriting when metadata appears both above and interleaved within the numeric matrix.

The final packaging step constructs a specmine-compatible dataset object containing the spectral matrix, sample metadata table, x-axis labels, spectroscopic modality type, and axis label text. Residual NA rows arising from undetected header lines are removed at this stage via a complete-cases filter applied jointly to the x-axis vector and the spectral matrix. The output is exported in two formats: an RDS file preserving the complete dataset object for use with the specmine R package, and a CSV file for integration with the webspecmine web application.

3.2 Supported Formats and Modalities

The pipeline supports input files in four formats: comma-separated values (CSV), tab-separated plain text (TXT), tab-separated values (TSV), and the proprietary binary SPC format used by GRAMS/AI and Thermo Scientific instrumentation. Format validation is performed as the first diagnostic step, with unsupported extensions causing the pipeline to halt with an informative error message prior to any further processing. For text-based formats (CSV, TXT, TSV), the pipeline imposes no structural requirements beyond the presence of a numeric spectral matrix. Metadata blocks, column headers, delimiter type, decimal separator, and matrix orientation are all determined automatically by Parser 1, without user intervention. This design accommodates the wide structural heterogeneity observed in openly available spectroscopic datasets, where formatting conventions vary across instruments, software packages, and research groups.

For SPC files, parsing is delegated to the hyperSpec package, which handles the binary structure of the format natively. Files with non-standard encoding such as those containing UTF-16 byte order marks are not supported by the current version of hyperSpec and are handled gracefully by the pipeline through a tryCatch wrapper, which intercepts the resulting error and returns an informative message without interrupting execution.

Three spectroscopic modalities are supported: Infrared (IR), Raman, and Ultraviolet-Visible (UV-Vis). The modality is not inferred automatically but is instead provided by the user at runtime via an interactive prompt, ensuring correct assignment of axis labels and specmine type identifiers (ir-spectra, raman-spectra, uvv-spectra). These identifiers correspond directly to the type schema enforced by the specmine package.

The pipeline additionally supports datasets distributed across multiple individual files a common pattern in experimental workflows where each sample is stored as a separate two-column file containing the spectral axis and intensity values. In this case, files are placed in a designated input directory and automatically concatenated into a single matrix prior to processing. For Raman datasets, linear interpolation onto a common reference axis is applied during concatenation to accommodate differences in spectral step size between files, using the approx function from base R. For IR and UV-Vis datasets, concatenation is performed without interpolation, preserving the original

spectral values strictly; files with a differing number of spectral points are excluded with a warning.

3.3 Dataset Acquisition

To evaluate the pipeline's ingestion and harmonization capabilities, a total of seven real-world open-access datasets were selected. These datasets were sourced from diverse repositories, including Zenodo and Recherche Data Gouv, intentionally encompassing a wide array of spectroscopic techniques (UV-Vis, IR, and Raman) and structural morphologies. The selected datasets reflect typical experimental heterogeneity, ranging from low-resolution mineralogical readings to high-throughput metabolomic screens[31,35-40].

3.4 Validation Strategy

To ensure computational robustness and logical accuracy of the ETL pipeline, a two-tiered validation strategy was employed.

The first tier consisted of localized unit testing using synthetic, deliberately corrupted, and severely unformatted datasets. This phase was designed to stress-test the diagnostic parser (Parser 1) and the execution module (Parser 2) in isolation, verifying the accurate triggering of individual flags, such as metadata extrusion, transposition correction, and matrix contiguousness restoration.

The second tier of validation focused on interoperability and structural integrity using the seven real-world datasets. Files that were successfully ingested, harmonized, and exported by the pipeline as RDS objects were directly loaded into the specmine computational ecosystem. Validation was considered successful if the objects passed the internal `check_dataset` function without errors, confirming that the dynamic experimental matrices were correctly packaged and inherently compatible with advanced machine learning environments.

3.5 Variability Analysis

Beyond structural compatibility, it was imperative to verify that the harmonization process preserved the biological, physical, and chemical integrity of the raw data. To quantify this, a localized statistical variability analysis was conducted using the specmine package. Each sample within a dataset was mathematically compared against the mean consensus spectrum of its respective group using three distinct metrics, each serving a specific analytical purpose:

Mean Absolute Error (MAE): Utilized as a linear magnitude error metric, MAE calculates the point-to-point absolute difference between a sample and the mean spectrum.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |x_i - \mu_i|$$

Where n is the total number of spectral points, x_i is the signal intensity of the individual sample at point i , and μ_i is the signal intensity of the mean consensus spectrum at point i . This metric provides a direct, easily interpretable quantification of routine biological or instrumental variability (signal stability) without exponentially punishing minor baseline shifts.

Mean Squared Error (MSE): Acting as a critical anomaly detector, MSE squares the spectral differences, thereby exponentially penalizing extreme deviations.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (x_i - \mu_i)^2$$

Where the variables follow the same notation as Equation 1. While MAE registers soft deviations, MSE is highly sensitive to physical degradation, severe light scattering, or corrupted peaks, allowing for the automated isolation of critical outliers.

Kullback-Leibler (KL) Divergence: Unlike MAE and MSE, which evaluate amplitude magnitude, KL Divergence operates as a shape metric, calculating the distance between spectral probability distributions.

$$D_{\text{KL}}(P \parallel Q) = \sum_{i=1}^n P(i) \log \left(\frac{P(i)}{Q(i)} \right)$$

Where $P(i)$ represents the normalized probability distribution of the sample spectrum, and $Q(i)$ represents the probability distribution of the reference mean spectrum. This parameter was selected for its chemical sensitivity; it detects changes in spectral morphology (such as the emergence of new oxidation peaks) independently of signal intensity. Consequently, KL Divergence effectively differentiates between simple physical contamination and true chemical evolution within the matrix.

4 Results and Discussion

4.1 IR Analysis

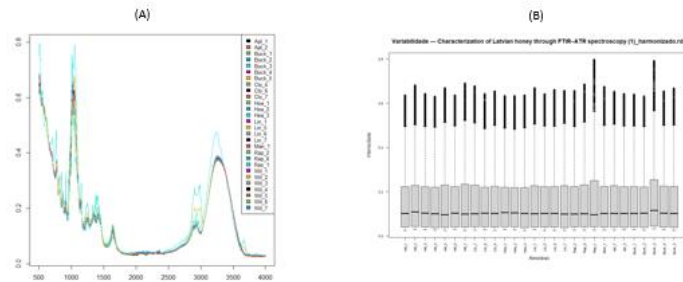


Figure 1. Spectral and variability analysis of Latvian honey. (A) Infrared spectral profiles representing high structural homogeneity across a high-density dataset. (B) Global dispersion of signal intensities, confirming baseline stability and highlighting *buck:3* as the primary source of minor intra-group variation.

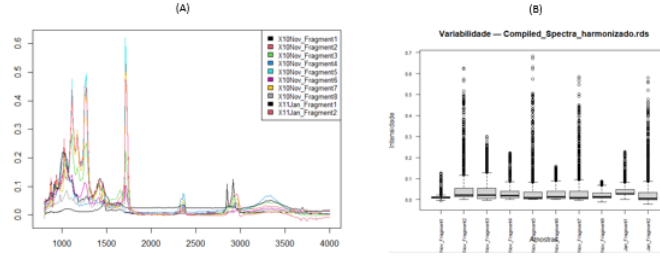


Figure 2. Spectral and variability analysis of composting plastic fragments. (A) Infrared spectra of macroscopic plastic fragments recovered from industrial composting, exhibiting severe environmental degradation. (B) Global dispersion of signal intensities, confirming the preservation of characteristic high-intensity absorbance peaks.

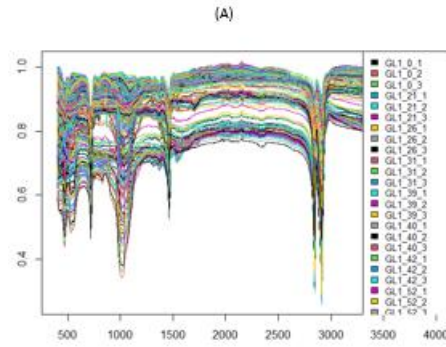


Figure 3. Spectral and variability analysis of plasticulture films degradation. (A) Infrared spectra of plastic materials subjected to UV treatment and physical abrasion, demonstrating high intra-group variability.

The Infrared (IR) datasets provided contrasting scenarios of structural homogeneity and severe environmental degradation. The high-density Latvian honey dataset demonstrated remarkable baseline stability and structural homogeneity, yielding minimal MAE deviations, with the *buck:3* sample identified as the primary source of minor intra-group variation. In stark contrast, the environmental datasets macroscopic composting plastic fragments and plasticulture films subjected to UV treatment and physical abrasion exhibited high intra-group variability and severe degradation. However, even in these highly heterogeneous datasets, the mathematical evaluation confirmed the preservation of characteristic high-intensity absorbance peaks, proving the robustness of the pipeline.

4.2 Raman Analysis

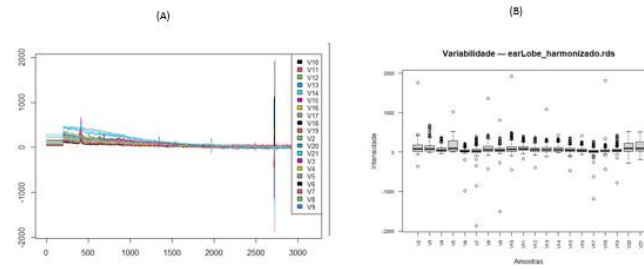


Figure 4. Spectral and variability analysis of zircon radiation damage (Raman). (A) Raman spectra of the dataset. The x-axis represents the Raman shift (cm^{-1}) and the y-axis indicates the spectral intensity. Each colored line represents a distinct mineral sample. (B) Global dispersion of spectral intensities across individual samples. The boxplots illustrate the intra-sample data distribution

The evaluation of the Raman spectroscopy pipeline utilized a dataset focused on zircon radiation damage. The analysis mapped the Raman shift and spectral intensity across multiple distinct mineral samples. Through the evaluation of global dispersion and intra-sample data distribution, the pipeline demonstrated the ability to handle the inherent structural complexities of the samples. In this context, the linear magnitude error metric (MAE) served as a direct quantification of instrumental and biological variability, confirming the overall signal stability and consistency of the interpolated profiles prior to extreme anomaly detection.

4.3 UV-Vis Analysis

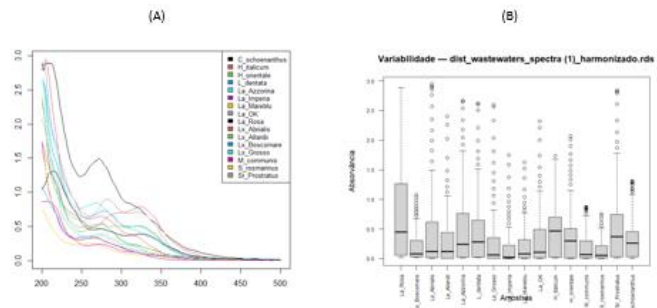


Figure 5. Spectral and variability analysis of aromatic plant distillation wastewaters. (A) UV-Vis absorbance spectra demonstrating morphological heterogeneity. (B) Global dispersion of absorbance intensities across individual samples; the boxplots visually highlight *La_Rosa* as an outlier.

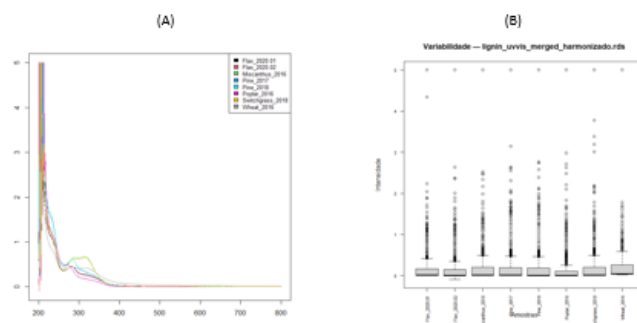


Figure 6. Spectral and variability analysis of biomass and lignin. (A) UV-Vis absorbance spectra displaying high spectral heterogeneity among the studied species. **(B)** Global dispersion of spectral intensities, confirming signal stability post-ingestion despite natural variation.

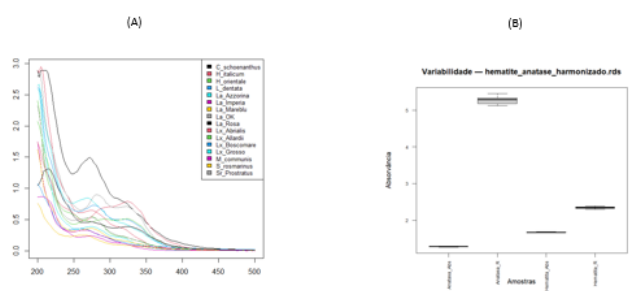


Figure 7. Spectral and variability analysis of hematite and anatase minerals. (A) UV-Vis spectra capturing the morphological profiles from different acquisition techniques (absorbance vs. reflectance). **(B)** Global dispersion of spectral intensities, reflecting the strong variability between the mineral classes.

The UV-Vis modality was evaluated using datasets with pronounced morphological heterogeneity, including aromatic plant distillation wastewater, biomass and lignin extracts, and hematite and anatase minerals. The distillation wastewater exhibited diverse spectral profiles, with the global dispersion analysis identifying the *La_Rosa* sample as a clear outlier. Similarly, the biomass and lignin dataset displayed high spectral heterogeneity among the studied species. Despite this natural variation, the harmonization process successfully preserved signal stability post-ingestion, which is quantitatively reflected by low Mean Absolute Error (MAE) values across the samples. Furthermore, the hematite and anatase dataset demonstrated strong variability between the mineral classes, accurately capturing the distinct morphological profiles generated by different acquisition techniques.

5 Conclusions

The advancement of integrative multi-omics heavily depends on the standardization and accessibility of analytical data. While spectroscopic techniques such as Infrared (IR), Raman, and UV-Vis are fundamental for high-throughput metabolic profiling, their integration has been historically hindered by data fragmentation and the absence of domain-specific metadata frameworks. This work successfully addresses this critical infrastructure gap through the development of a tailored Extract, Transform, and Load (ETL) pipeline, establishing the foundation for a centralized, FAIR-compliant spectroscopic repository. The proposed architecture demonstrated robust data harmonization capabilities across seven highly heterogeneous real-world datasets. By automating format detection, metadata extrusion, and matrix normalization without relying on heavy external dependencies, the pipeline effectively bridged the structural disparities inherent to open-access spectroscopic data. The rigorous dual-tiered validation strategy confirmed that the harmonization process maintained strict structural compatibility with the specmine computational ecosystem. Furthermore, the statistical variability analysis utilizing Mean Absolute Error (MAE), Mean Squared Error (MSE), and Kullback-Leibler (KL) divergence proved that the automated ingestion preserved the biological, physical, and chemical integrity of the raw spectra while successfully isolating critical morphological anomalies.

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Code Availability

All scripts and diagnostic parsers developed for this study are publicly available on GitHub at: <https://github.com/Progamigor/Projeto-Repository-of-Experimental-Datasets-for-Spectroscopic-Techniques.git>